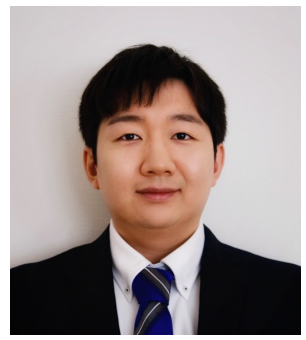
EMNLP2026 BUDAPEST

A method to efficiently
Adapt an LLM to
A language or domain.

OPTIMER

Optimal Distribution Vector Merging
Is Better than Data Mixing for Continual Pre-Training



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One Model Does Not Fit Every Language or Domain

- Thus, governments and companies adapt LLMs to

Local Languages
ASEAN languages, ...

Qwen3

Continual pre-train on
Southeast Asian languages



SEA-LION 4 Model
from **AI Singapore**

Domains
Medical, ...

LLM-jp

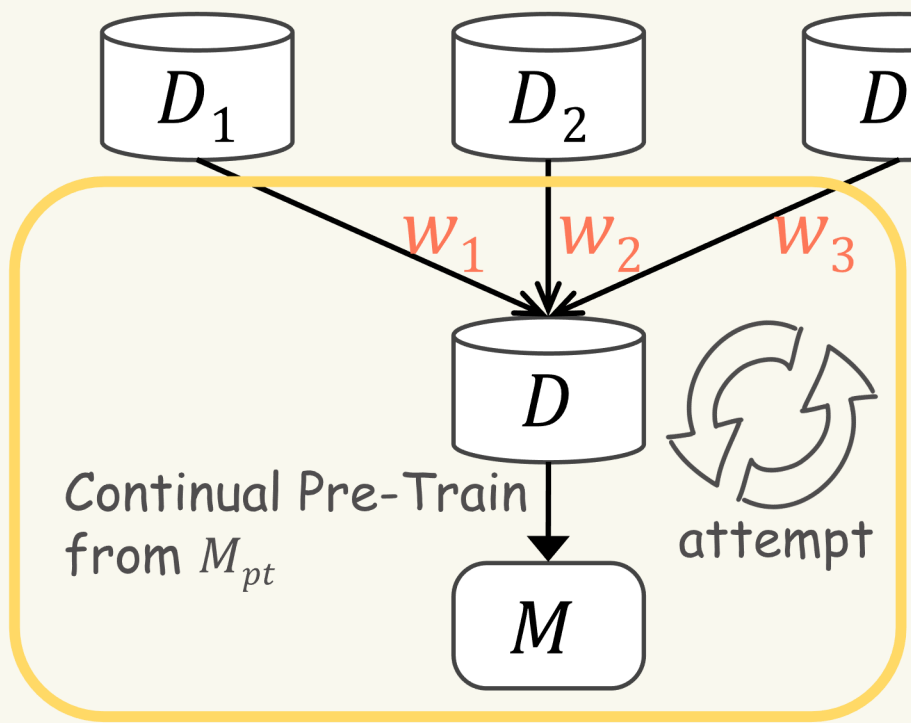
CPT on
Medical Data



Adapting LLMs is Costly

- A widely used method is continual pre-training with data mixing,
  - i.e. training LLM on target languages/domains data and English data.
- However, Data Mixing is costly because it requires multiple trials to search for the optimal **mixing ratio** of training datasets D_i.

(a) Data Mix



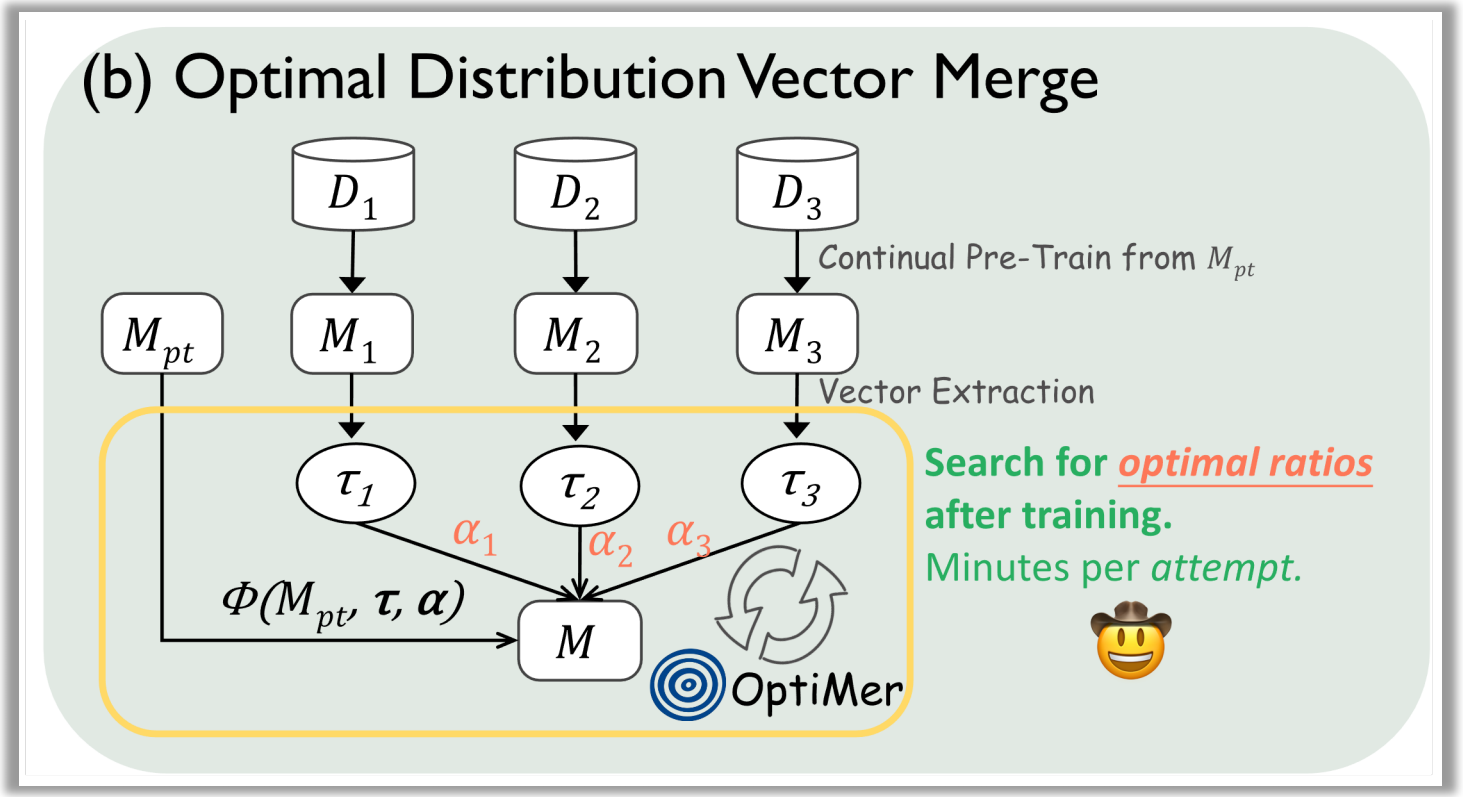
Fix *ratios* before training.
Weeks per *attempt.*
Suboptimal? Start over!

Motivation

- Training is expensive,
  so we do it only once.
- Finding the optimal mixing ratio
  by adjusting it many times,
  but each in minutes.

We Propose OptiMer

- Train separate models M_i from pre-trained model M_{pt}.
- Train on one dataset, e.g.
  - Japanese dataset D_1 -> M_1
  - English dataset D_2 -> M_2
  - Math dataset D_3 -> M_3



- Extract distribution vectors $\tau_i = M_i - M_{pt}$

- Search for the optimal vector combination through Bayesian optimization using the **Optuna Framework**.



Optuna-kun

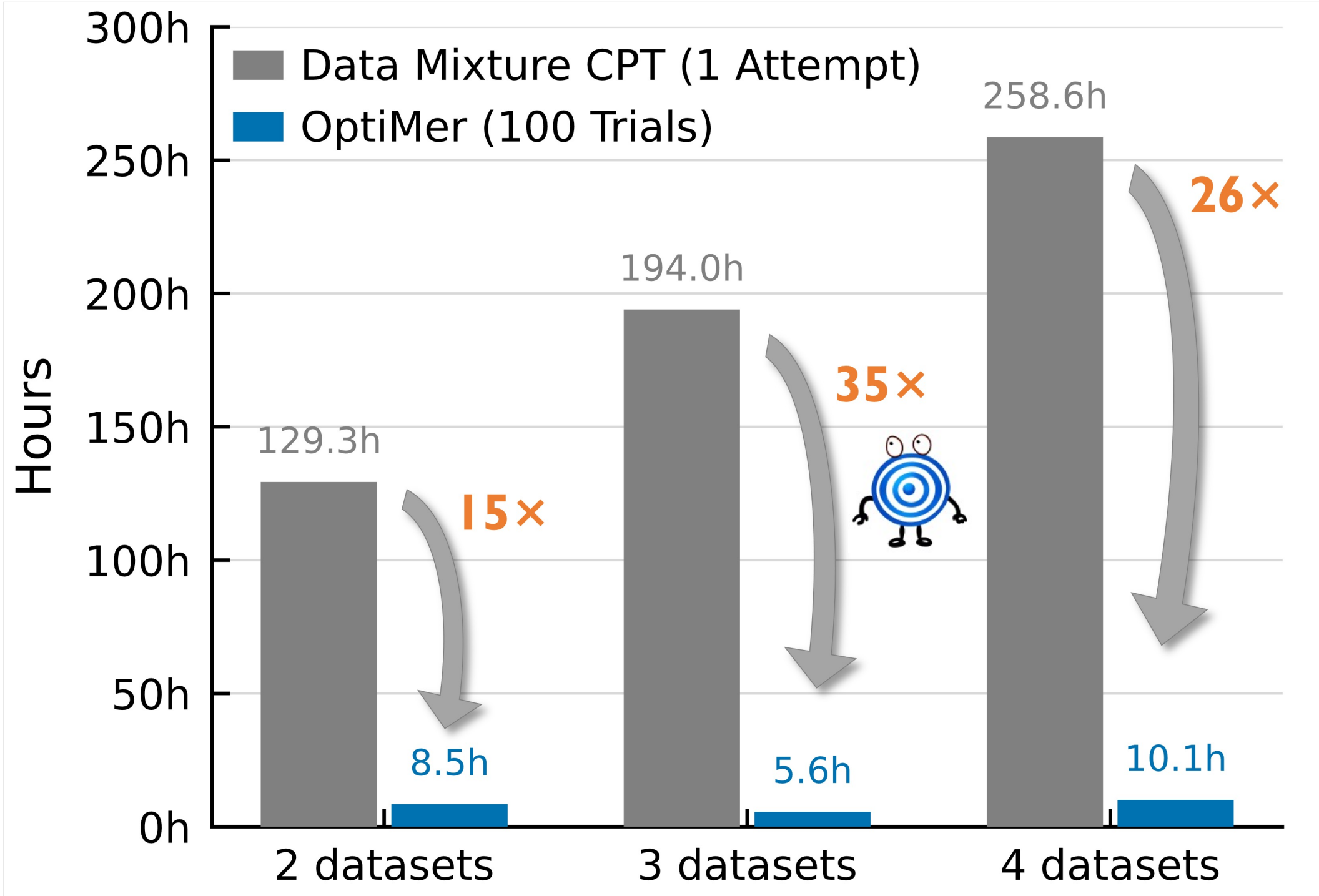
Experimental Settings

Model	Gemma 3 27B
Data	1B tokens each: Japanese · English · Chinese · Code · Math
Baselines	DataMix with equal ratio Average merging: Task Arithmetic, TIES, DARE
OptiMer	DARE-Linear merging Bayesian optimization, 100 trials
Evaluation	En / Ja / Zh / Math / Code benchmarks, 1-shot

Check our paper for detailed settings

OPTIMER is Efficient

- After the first round of training, our ratio optimization is much more efficient than data mixing.



OPTIMER is Effective

Combination	Method	English				Math	Code		Japanese	Chinese	Avg.
		MMLU	ARC-C	HellaSwag	TQA	GSM8K	HE	MBPP	JA LB	C-Eval	
	Gemma 3 27B PT	31.59	27.30	48.41	54.71	0.83	0.00	0.60	39.04	27.41	25.54
	Gemma 3 27B IT	51.03	34.90	55.91	64.87	15.39	2.44	37.80	54.64	37.96	39.44
	Single CPT, merged with IT (α_{it}=0.6)										
	CPT_{En}	73.60	69.80	82.82	46.39	79.98	71.95	68.40	43.06	62.56	66.51
	CPT_{Ja}	72.11	67.06	81.98	55.32	83.78	68.29	68.20	72.50	62.48	70.19
	CPT_{Zh}	72.85	68.94	81.70	48.84	82.18	71.95	67.80	57.98	62.85	68.34
	CPT_{Math}	75.07	70.14	82.72	43.57	82.64	57.32	68.00	72.79	63.37	68.40
	CPT_{Code}	71.86	70.05	81.71	51.16	81.96	64.63	68.40	73.39	62.26	69.49
Ja + Math	$\text{DataMix}_{\alpha_{\text{it}}=0.6}$ (Baseline)	73.17	68.77	81.72	49.57	83.62	50.00	68.80	73.34	61.74	67.86
	$\text{DataMix}_{\text{OptiMer ratio}}$ (Ours)	72.72	**70.05**	82.62	50.06	76.19	**67.07**	64.80	72.85	61.96	68.70
	Task Arithmetic (Ilharco et al., 2023)	**75.44**	69.88	83.31	41.49	74.45	60.37	1.60	**75.11**	**63.67**	60.59
	TIES (Yadav et al., 2023)	71.14	65.70	80.03	30.11	69.52	26.22	0.00	72.40	58.84	52.66
	DARE-Linear (Yu et al., 2024)	75.02	69.71	**83.74**	42.96	78.17	58.54	65.00	72.37	63.30	67.65
	DARE-TIES (Yu et al., 2024)	74.72	68.94	82.75	39.53	73.77	60.98	0.80	74.39	62.33	59.80
	OPTIMER (Ours)	73.17	68.34	82.50	**55.20**	**84.46**	59.76	**70.40**	72.53	63.45	**69.98**
Ja + Code	$\text{DataMix}_{\alpha_{\text{it}}=0.6}$ (Baseline)	71.23	69.20	81.14	43.08	78.62	64.02	65.00	72.38	60.40	67.23
	$\text{DataMix}_{\text{OptiMer ratio}}$ (Ours)	**72.35**	68.60	80.22	51.77	74.00	55.49	67.60	**74.66**	**62.26**	67.44
	Task Arithmetic (Ilharco et al., 2023)	72.33	**70.39**	82.44	42.72	78.17	45.12	60.20	74.04	61.96	65.26
	TIES (Yadav et al., 2023)	66.44	66.98	79.69	36.11	70.58	43.29	53.60	70.22	54.75	60.18
	DARE-Linear (Yu et al., 2024)	72.33	70.31	**82.50**	42.84	79.08	46.34	60.60	74.07	62.18	65.58
	DARE-TIES (Yu et al., 2024)	71.14	69.37	82.00	41.49	76.04	42.68	60.40	73.30	60.25	64.07
	OPTIMER (Ours)	71.41	66.89	81.44	**54.22**	**83.47**	**64.63**	**70.20**	73.06	61.81	**69.68**
Ja + Zh + Math	$\text{DataMix}_{\alpha_{\text{it}}=0.6}$ (Baseline)	73.83	69.03	80.54	37.70	79.61	32.32	65.40	72.77	62.18	63.71
	$\text{DataMix}_{\text{OptiMer ratio}}$ (Ours)	74.34	67.58	81.80	44.80	81.27	66.46	67.20	**74.05**	60.48	68.66
	Task Arithmetic (Ilharco et al., 2023)	**75.27**	**69.97**	83.18	38.92	73.54	45.73	0.40	73.17	**64.41**	58.29
	TIES (Yadav et al., 2023)	70.13	66.72	79.65	26.68	65.81	14.63	0.20	69.86	60.03	50.41
	DARE-Linear (Yu et al., 2024)	74.48	69.28	**83.65**	38.68	76.42	50.00	62.00	70.92	63.89	65.48
	DARE-TIES (Yu et al., 2024)	74.06	69.54	82.59	35.86	70.36	45.12	0.40	72.38	63.60	57.10
	OPTIMER (Ours)	72.94	68.60	82.98	**51.77**	**83.93**	**67.07**	**70.20**	72.65	63.22	**70.37**

OptiMer is Flexible

- OptiMer **quickly adapts an LLM to another domain** through **re-optimization**.
  - Using the already calculated distribution vector set: Ja, Zh, En, Math
  - We can set another goal, and only re-optimize the vector merging ratios **in several hours**, **without training the whole model again**!

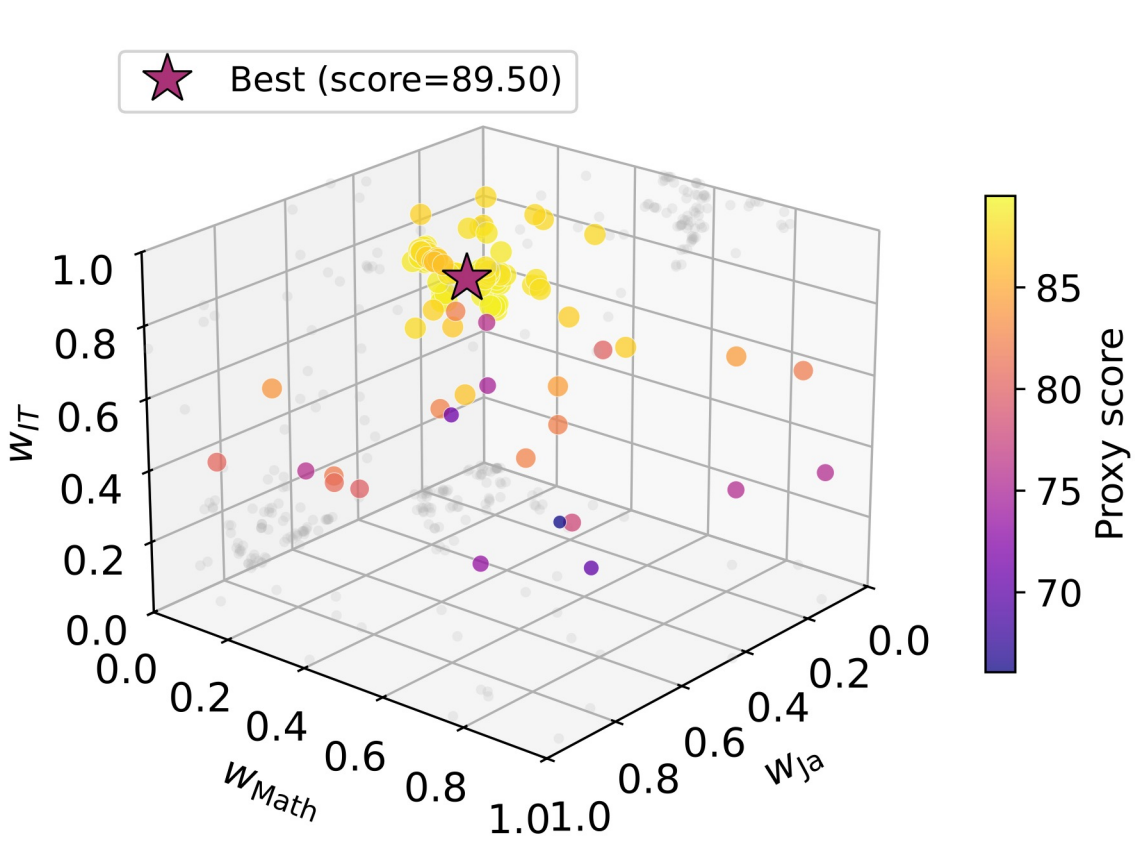
Objective	English				Math	Code		Japanese	Chinese	Avg.
	MMLU	ARC-C	HellaSwag	TQA	GSM8K	HE	MBPP	JA LB	C-Eval	
English	72.87	**69.62**	82.02	**47.74**	82.49	44.51	71.40	69.56	62.93	65.35
Math	70.10	63.05	79.16	41.13	82.87	4.88	**72.60**	68.47	58.99	58.37
Japanese	**73.96**	68.86	**82.73**	46.39	**84.69**	**68.29**	69.20	**73.37**	62.85	**69.07**
Chinese	71.93	69.11	82.37	34.76	72.71	11.59	1.60	40.78	**63.60**	44.29

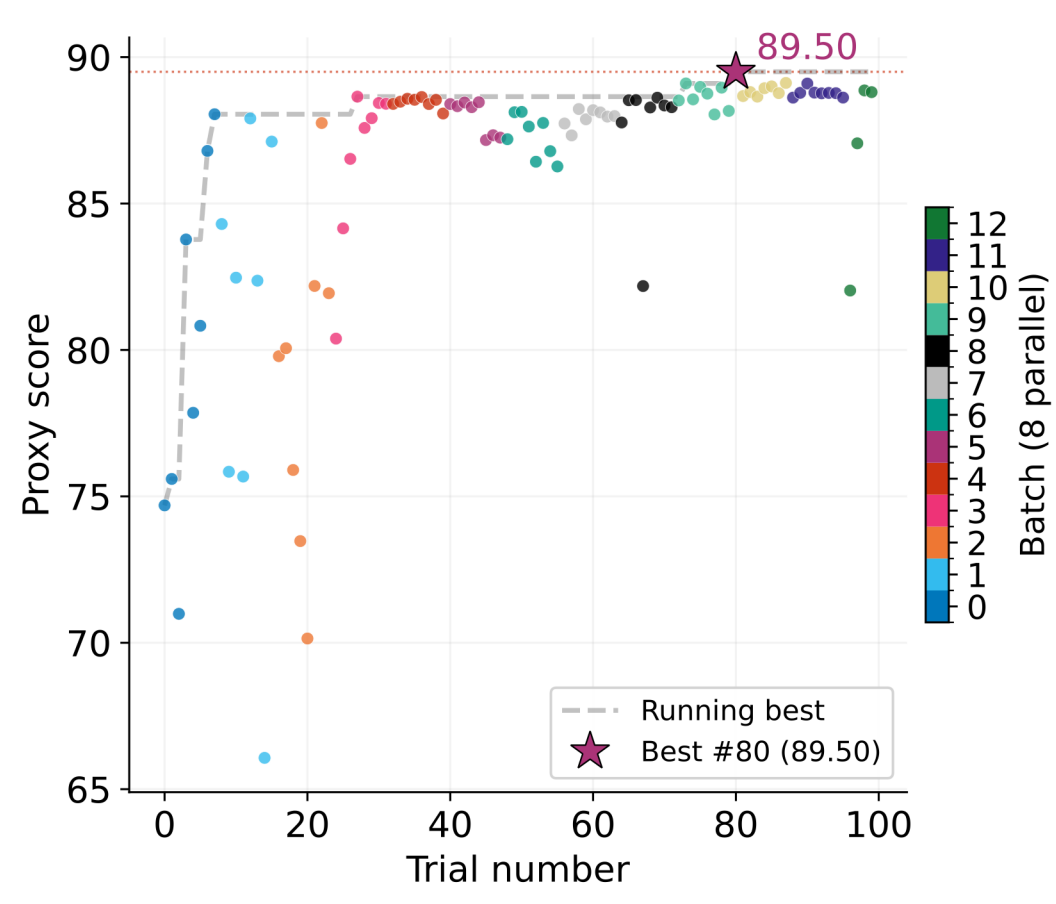


OptiMer's Dynamics

- Good ratio combinations are concentrated.
- Usually with large IT and small CPT weights.
- Thus, grid search is impractical.

- OptiMer quickly approaches this good region, usually converging within 100 trials.



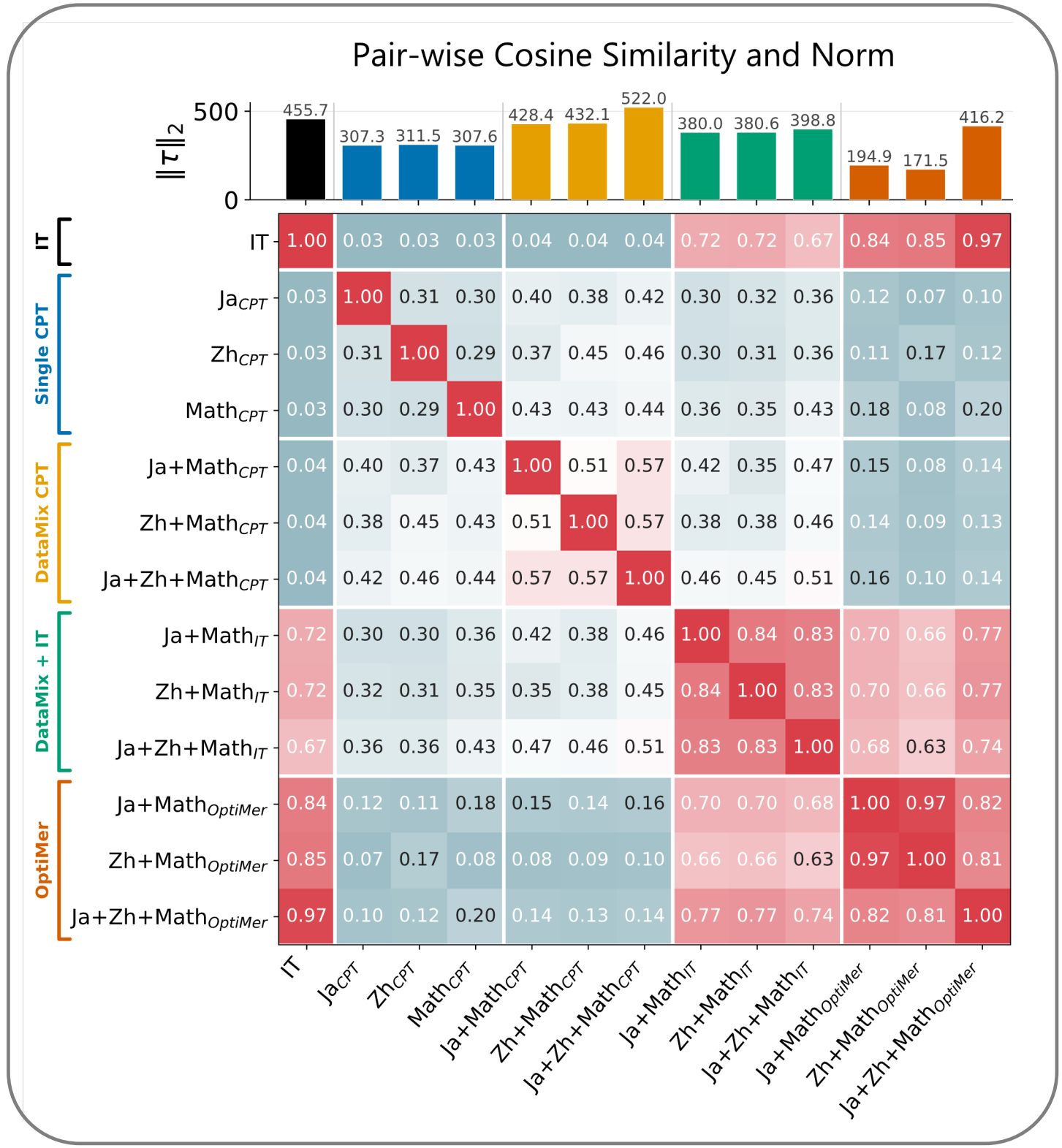


Why Mergeable?

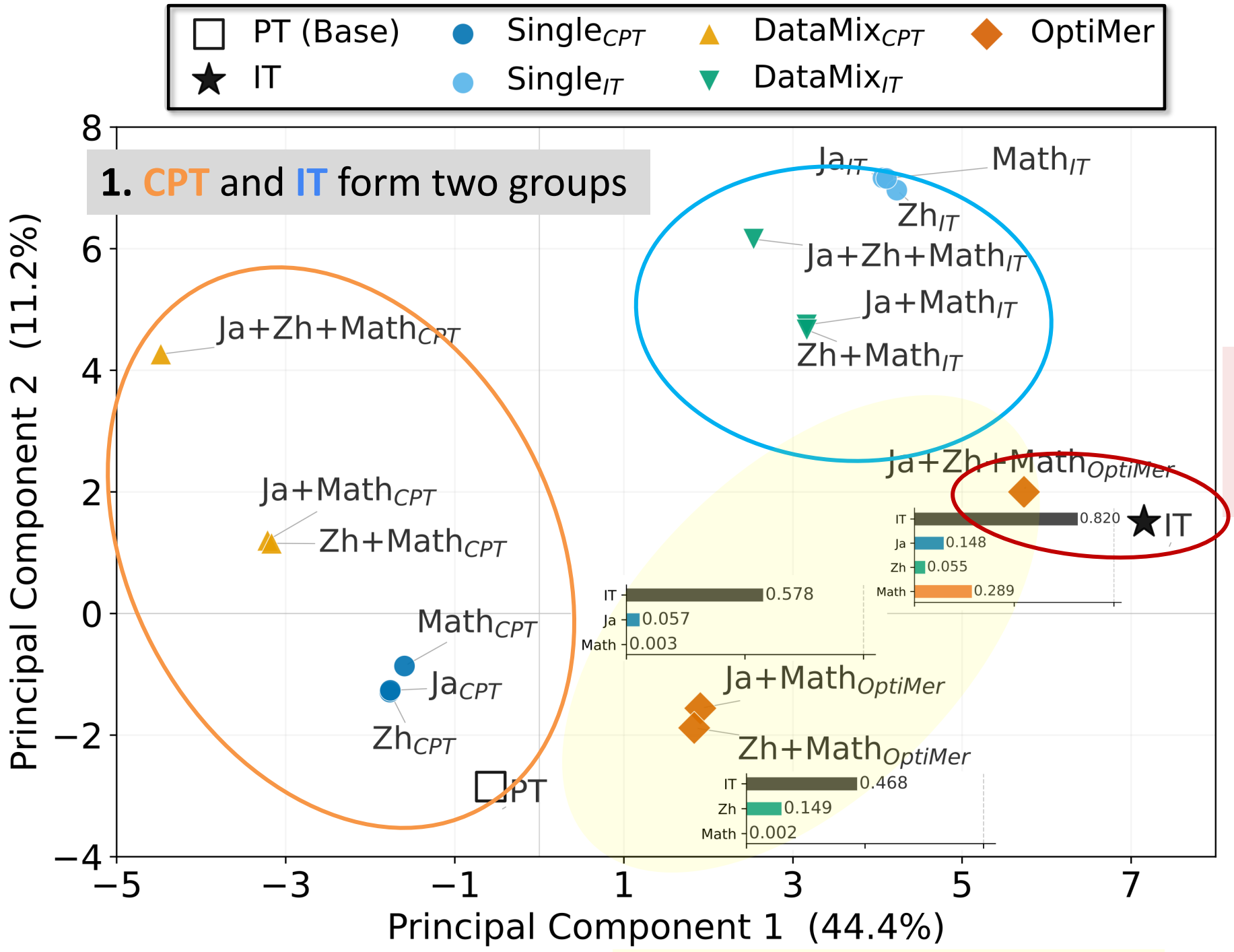
Distribution vectors are nearly orthogonal.

- CPT and IT vectors are **nearly orthogonal** 0.03.
- Different CPT vectors also exhibit **low similarity** 0.29–0.31.

→Vectors are mergeable with low conflicts.



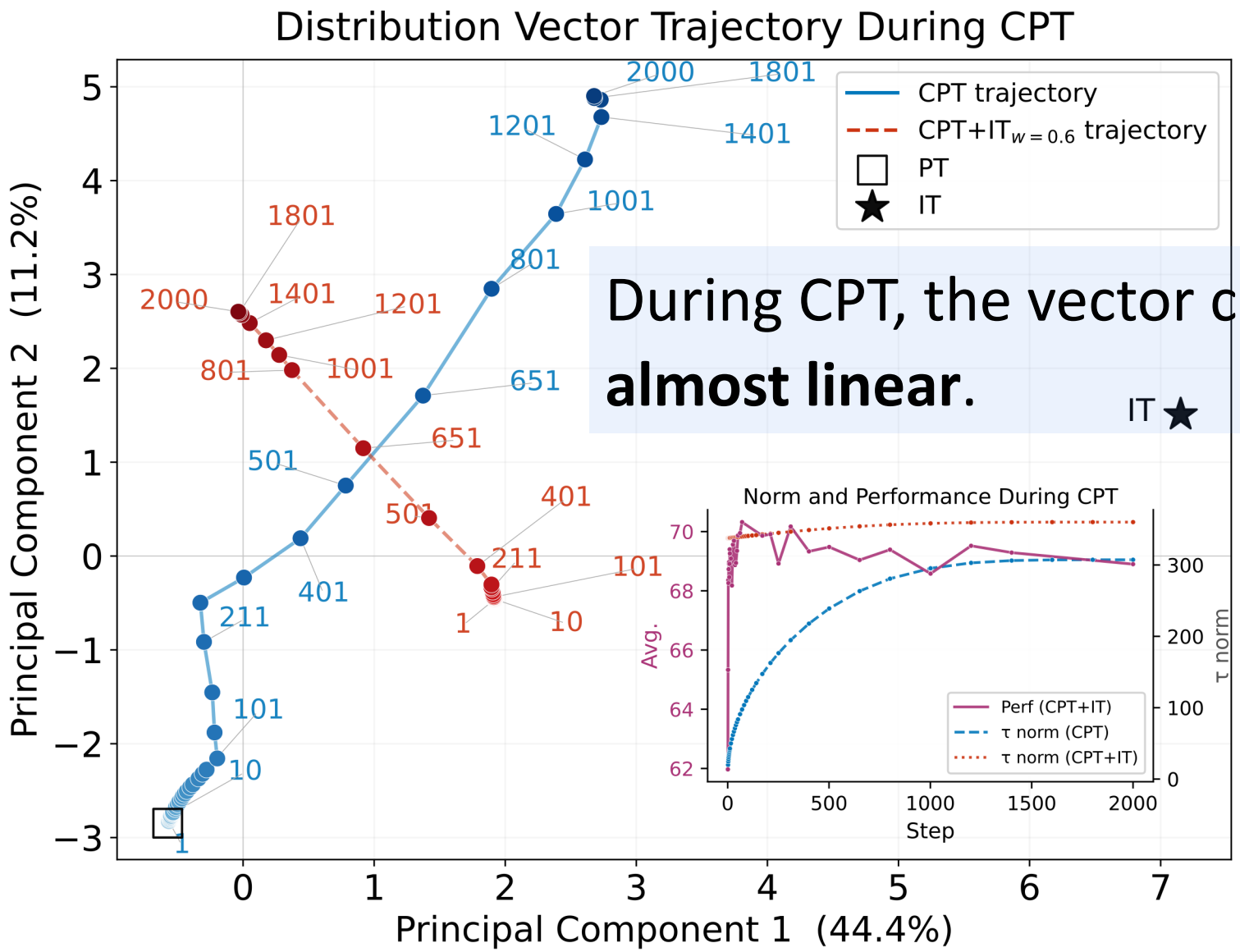
Distribution Vector PCA



3. The best model is actually near Gemma-IT

2. The three OptiMer models form another group

Continual Pre-Training Dynamics



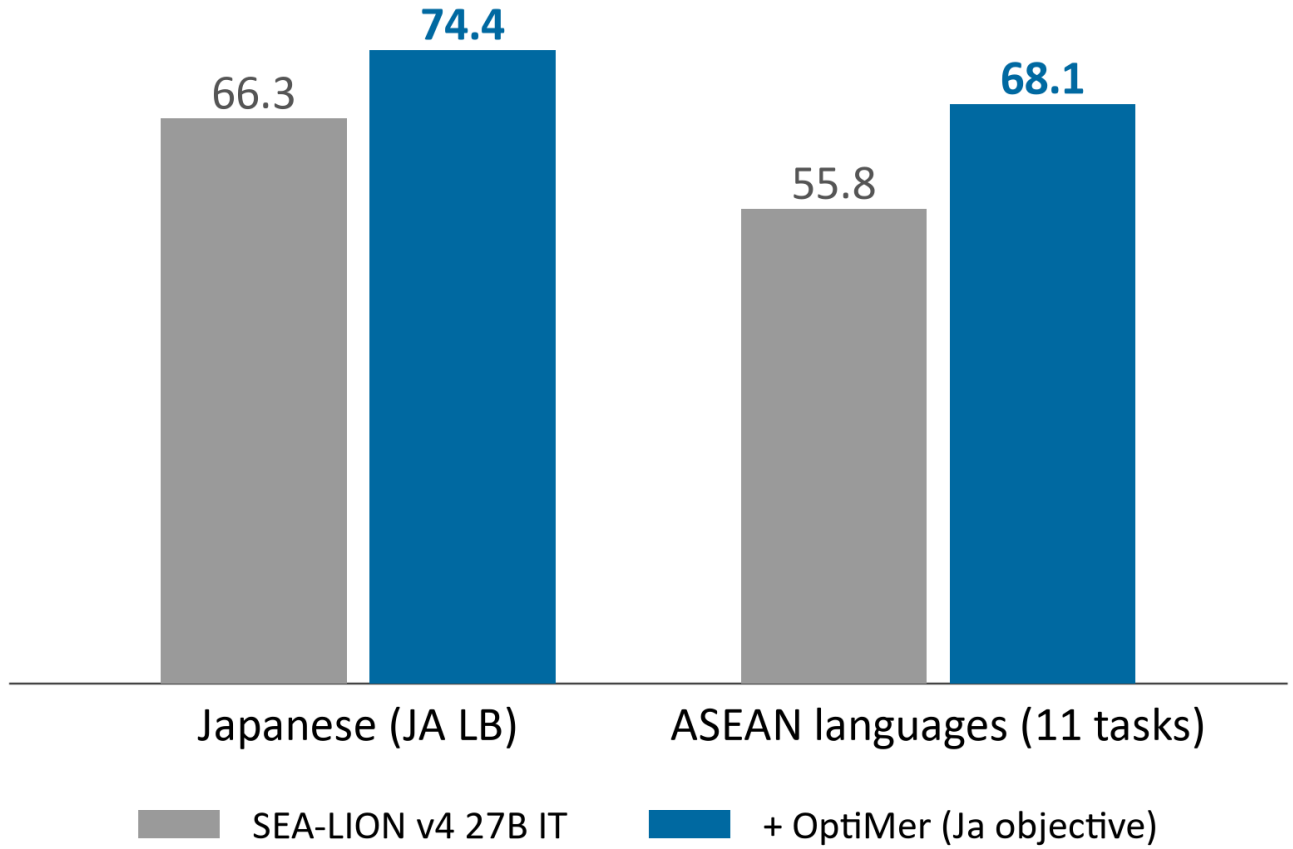
During CPT, the vector change is **almost linear**.

- It suggests that controlling the **vector merging ratio** has similar effect of controlling the **training duration**.

Applications

- We applied OptiMer to SEA-LION-v4-27B-PT, a model for ASEAN languages, to further enhance its Japanese capability, using distribution vectors from: Japanese, Chinese, English, ASEAN languages, Math, Code

- Performance on Japanese tasks is improved from 66.34 to 74.40 (+8.1), and ASEAN tasks also improve, from 55.76 to 68.07.



Take-Home Message

- **OPTIMER** adapts open-sourced LLMs
- to Your language and Your domain,
- efficiently, effectively, and flexibly.

Thank you for your attention!

